Gravitational potential energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate height change to near-surface gravitational energy.

SUCCESS CRITERIA - I can...

1. Choose a height reference.
2. Use ΔU=mgΔh with the stated g.
3. Distinguish potential energy from force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “potential energy” and “reference level” to explain your model or result. Sentence starters are optional.

potential energy: energy associated with the configuration of interacting objects

reference level: the chosen position where potential energy is set to zero

weight: the gravitational force on an object

01 Warm-up: bring back an earlier idea

A cart’s kinetic energy changes from 15 J to 3 J.
Find net work and state whether its speed rises or falls.

Word help: kinetic energy = energy associated with motion

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

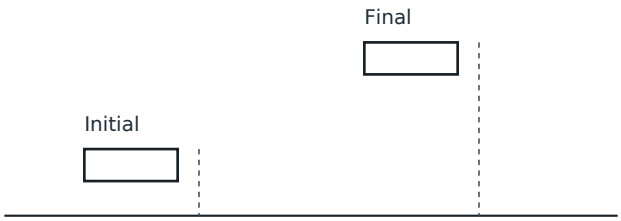
02 Worked example: explain the model

Reference: $\Delta U = mg\,\Delta h = mg(h_f - h_i)$.
Approximately constant g = 10 N/kg. Choose a zero height; the energy change does not depend on that choice.

Worked reasoning: Raising a 2 kg object by 3 m with g=10 N/kg increases gravitational potential energy by mgh=60 J. The chosen zero does not affect this change.

Your turn: Draw initial and final height levels and label the 3 m difference.

Sentence starter: Both ... . They differ in ... because ... .



Choose and label a zero-height level.

Marzano check - circle: ① ② ③

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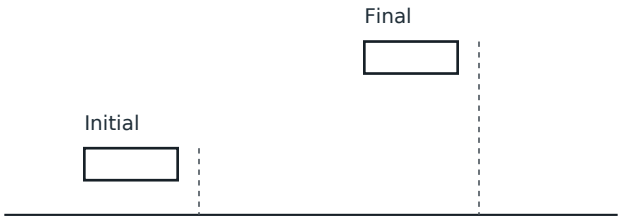
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two identical books sit at heights 1 and 3 m above the floor. Which has more gravitational potential energy relative to the floor?

Sentence starter: I notice … . This supports … because … .



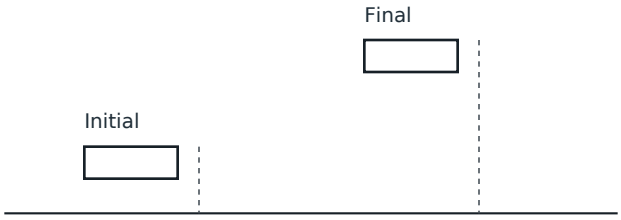
Choose and label a zero-height level.

Marzano check - circle: ① ② ③

04 Apply the physics

Each book in question 3 has mass 2 kg.
Use g = 10 N/kg and U = 0 at the floor.
Find U at 1 m and at 3 m. Then find ΔU when one book is lifted from 1 m to 3 m.
Mark the vertical change.

Sentence starter: I found … [unit]. This value means … in this situation.



Choose and label a zero-height level.

Marzano check - circle: ① ② ③

05 Explain your reasoning

Does a negative value of U relative to a chosen zero imply a negative mass or impossible height?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Equal-mass carts have velocities +6 and -4 m/s before an ideal 1 D elastic collision.
Give both final velocities.
Use before and after snapshots to organize the momentum account.

Word help: elastic = a collision that conserves total kinetic energy

Sentence starter: I notice … . This supports … because … .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

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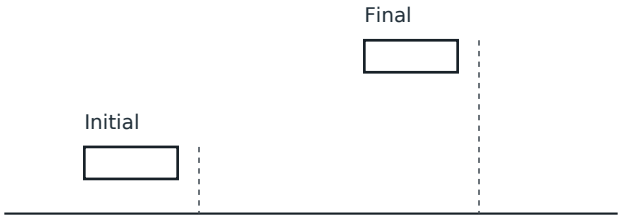
Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Lift a 3 kg object by 2 m.
Find ΔU with g=10.
Label the relevant heights or distances on the model.

Sentence starter: I found ... [unit]. This value means ... in this situation.



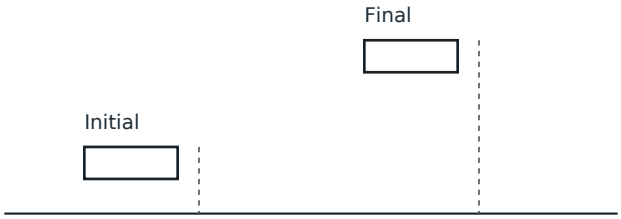
Choose and label a zero-height level.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Construct two mass-and-height-change pairs with the same 60 J increase using g=10.
Label the relevant heights or distances on the model.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Choose and label a zero-height level.

Marzano check - circle: 1 2 3